

Lem subgrupo automorfismos fijan

Lema 1. Sea $\mathfrak{A} = (A, z^{\mathfrak{A}})$ una L-estructura. Entonces, para un subconjunto $B \subset A$, $\text{Aut}(A/B)$ es un subgrupo de $\text{Aut}(A)$.

Demostración: Verificamos que $\text{Aut}(A/B)$ es un subgrupo de $\text{Aut}(A)$:

- (i) No vacío: $\text{id}_A \in \text{Aut}(A/B)$, pues $\text{id}_A(b) = b$ para todo $b \in B$.
- (ii) Clausura: si $h, g \in \text{Aut}(A/B)$, para todo $b \in B$ se tiene $(g \circ h)(b) = g(h(b)) = g(b) = b$, luego $g \circ h \in \text{Aut}(A/B)$.
- (iii) Inversos: si $h \in \text{Aut}(A/B)$, para todo $b \in B$ se tiene $h(b) = b$, luego $h^{-1}(b) = b$, y por tanto $h^{-1} \in \text{Aut}(A/B)$. ■

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